

Measuring and Mitigating Alignment Risks of Language Models

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My research studies how language model systems might create and amplify long-term risks through interactions with people, and develops methods to measure and mitigate these socio-affective risks. Most current language model “alignment” methods make simplifying assumptions about this problem: that human preferences can be aggregated into a single reward, that undesirable behavior can be caught by static evaluations specified in advance, and that behaviors of model’s or user’s can be sufficiently evaluated at a single point in time. These assumptions become increasingly fragile as language model interactions extend to diverse users or more complex and interpersonal situations, but also become more dangerous as models are increasingly—and sometimes unknowingly—integrated into people’s lives.

I focus in particular on a few complementary scenarios. One is on model failures that emerge in contexts where human preferences, values, and social behavior conflict with a model’s alignment objects, leading to behaviors such as sycophancy, manipulation or the suppression of minority perspectives. Another is when repeated human-model interactions (e.g., through chat interfaces) over very long periods of time shift and influence long-term human behavior. This could manifest through persuasion or polarization of beliefs, decline of cognitive abilities, or general behavioral changes to a human as a function of their interaction with models. Lastly, when existing evaluations are too simple to catch complex model behaviors, and adversarial or multi-agent systems are needed to automatically discover emerging model failures. Concretely, my work so far has been organized around three aspects of these problems.

- How can we evaluate and understand risks as they emerge in real and long-term interactions between humans and language models? [TBR: gpt-bias, tailoring, dices]
- How can we design alignment objectives for language models that better represent human values and preferences? [TBR: sct-rewards, values, intransitivity]
- How can we anticipate and robustly evaluate emerging harms of increasingly complex systems *before* they emerge when deployed to humans? [TBR: leagues, probing, sycophancy]

The first question asks how we can collect data for a better understanding of language model interactions. Safety evaluations are only as good as the behaviors they measure and the populations of people they measure this in. My work here has designed studies that collect human-AI interactions *longitudinally*, allowing for the measurement of what repeated exposure to models does to humans in the very long term, even if those effects are not apparent in real time[?]. We also show how collecting and evaluating models on data from diverse populations shows how significantly judgements can differ and how this can be incorporated in to improve models [?].

My second line of work asks how the reward structures and training objectives themselves should change, to allow us to build more robust alignment pipelines. Most prevalent methods for aligning language models aggregate conflicting preferences, values, or judgements into one objective or reward that guides model training—thus potentially masking the differences in humans that are important for fairness. We train language models with reward aggregation structures inspired by

utility theory, allowing for a better tradeoff between helpful and harmless behavior [?]. We fine-tune models specifically to adapt to differing values of individual users, allowing for systems that are more pluralistic and aligned [?].

Finally, many safety failures cannot be captured by evaluations specified in advance: we often do not know which behaviors or risks to test for until models exhibit them. Moreover, interactions in real-time might not show any signs of being harmful, but over long periods of sustained interactions with humans, could lead to risks to humanity. My third line of work develops methods to address this. We train populations self-improving language models whose purpose is to red-team (i.e., generate adversarial prompts to discover safety failures) the main language model being trained [?]. We probe the internal mechanisms through which such undesirable behaviors emerge and show that we can control it at inference time with activation-level mitigations [?].

Working at the frontier of deploying language model products has exposed me to a class of alignment problems that become visible only when models reach a sufficient bar of capability, are widely used, and can be projected to have continued longitudinal risks. We identify subtle ways in which language model generations can influence human behaviors, both in short interactions and along very long time scales [gpt-bias]. We detect and control for sycophantic behaviors of models—situations where preference optimization pipelines reward agreeableness and flattery—and how this can pose dangers to humans [control-syc]. We update the reinforcement learning reward formulation using social choice theory aggregation methods to allow for better aggregation of different rewards [sct-rlhf]. We show how intransitivity in human preferences can be better handled by alignment pipelines that handle game-theoretic objectives, and show how this can recover and handle erasure of minority preferences better. [intransitivity]. We create self-improving adversarial red-teaming systems to uncover dangerous capabilities of models [leagues]. We create datasets for evaluating differing safety perspectives, build models that can handle this better, and also work towards longitudinal measurements of data. [dices, pluralis, long-term]. Building off of this work, my future research will work on answering the following:

- How can alignment objectives be grounded in frameworks that allow for fairer representation of conflicting values and stakeholder preferences?
- How can we design self-improving multi-agent language systems that can surface emerging failure models and evolving risks in models?
- How can we measure and mitigate long-term risks through targeted controlled studies?

References